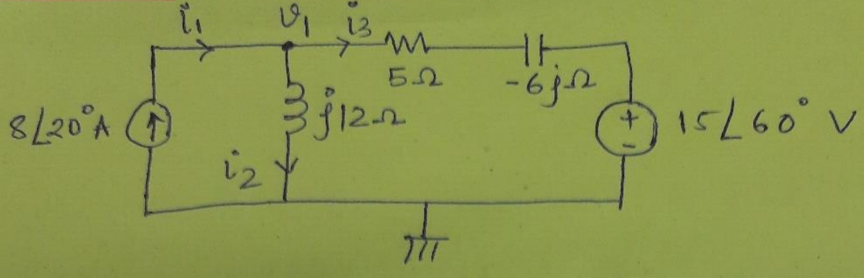
Nodal analysis for AC circuits :- (Example)



find the node voltage v_1 & branch currents i_1, i_2, i_3

Solution :- Using KCL at node v_1.

$$\bar{i}_1 = \bar{i}_2 + \bar{i}_3$$

$$8\angle 20^\circ = \frac{v_1}{j12} + \frac{v_1 - 15\angle 60^\circ}{(5 + (-6j))}$$

$$8\angle 20^\circ = v_1\left[\frac{1}{j12} + \frac{1}{5-6j}\right] - \frac{15\angle 60^\circ}{7.81\angle -50.194^\circ}$$

$$7.517 + j2.736 = v_1(-0.0833j + 0.0819 + 0.098j) - 1.9206\angle 110.194^\circ$$

$$7.517 + j2.736 = v_1[0.0819 + 0.0147j] + 0.663 - 1.8025j$$

$$v_1 = \frac{7.517 - 0.663 + j2.736 + 1.8025j}{(0.0819 + 0.0147j)}$$

$$v_1 = \frac{6.854 + 4.5385j}{0.0819 + 0.0147j} = 90.711 + 39.133j$$

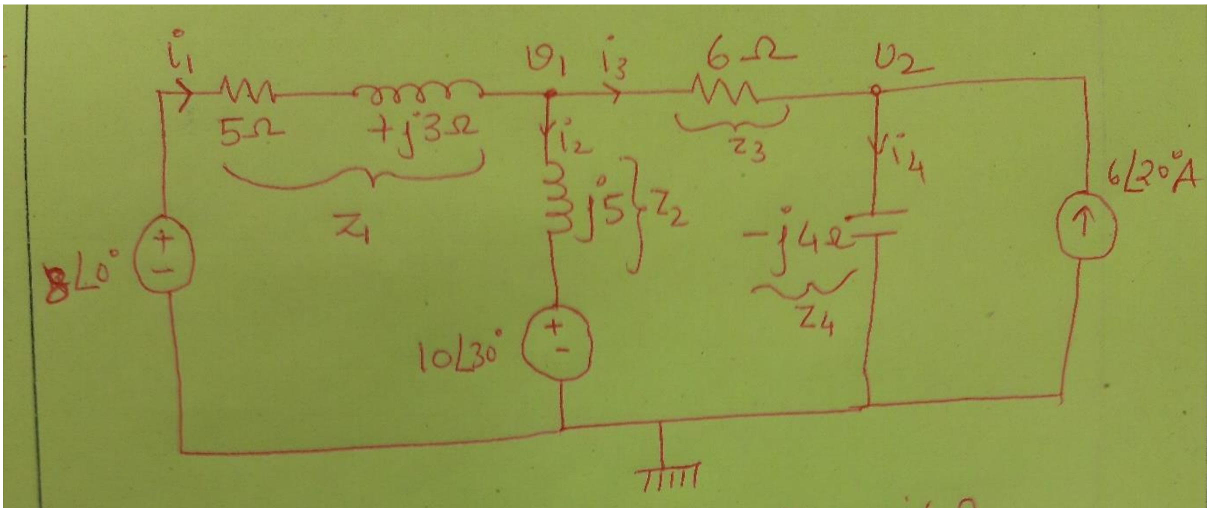
$$= 98.792\angle 23.33^\circ V$$

$$i_2 = \frac{v_1}{j12} = 3.260 + (-7.559)j = \underline{8.23\angle -66.67^{\circ} A}$$

$$i_3 = \frac{v_1 - 15\angle 60^{\circ}}{5-6j}$$

$$v_3 = 4.25 + 10.327j = \underline{11.167\angle 67.629} \text{ A}.$$

Find the node voltages V1 and V2.



$Z_1 = 5+j3\,\Omega,\ Z_2 = j5\,\Omega,\ Z_3 = 6\,\Omega,\ Z_4 = -j4\,\Omega.$

KCL at node v_1 —

$$\frac{8\angle 0^\circ - v_1}{Z_1} = \frac{v_1 - 10\angle 30^\circ}{Z_2} + \frac{v_1 - v_2}{Z_3}$$

$$\frac{8\angle 0 - v_1}{(5+j3)\Omega} = \frac{v_1 - 10\angle 30^\circ}{j5} + \frac{v_1 - v_2}{6}$$

$$\frac{8}{(5+j3)} + \frac{10\angle 30^\circ}{j5} = v_1\left[\frac{1}{5+j3} + \frac{1}{j5} + \frac{1}{6}\right] - \frac{v_2}{6}$$

$$1.1764 + (-0.705)j + 1 - 1.732j = V_1(0.5137 - 0.088j) - \frac{V_2}{6}$$

$$2.1764 - 2.437j = V_1(0.5137 - 0.088j) - \frac{V_2}{6} \quad \text{—} (1)$$

KCL at node V_2 -

$$\bar{I}_3 + 6\angle 20^\circ = i_4$$

$$\frac{V_1 - V_2}{6} + 6\angle 20 = \frac{V_2}{-j4}$$

$$\frac{V_1}{6} - V_2\left[\frac{1}{6} - \frac{1}{j4}\right] = -6\angle 20$$

$$\frac{V_1}{6} - V_2(0.166 + 0.25j) = -5.638 - 2.052j \quad (2)$$

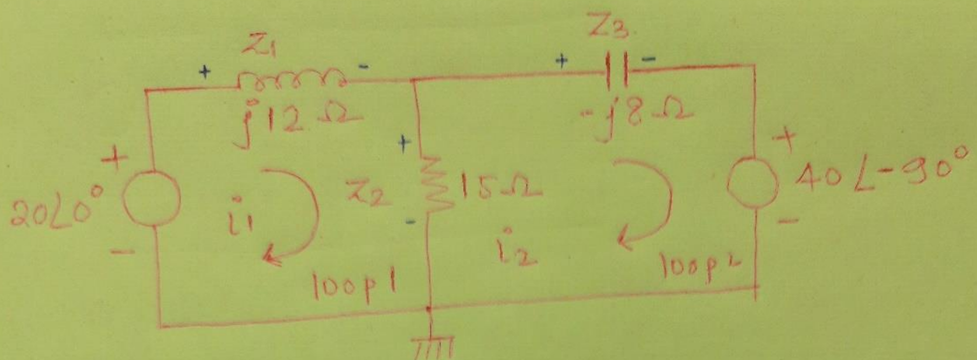
$$\begin{bmatrix} 0.5137 - 0.088j & -1/6 \\ 1/6 & -(0.166 + 0.25j) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2.1764 - 0.2437j \\ -5.638 - 2.052j \end{bmatrix}$$

Solve using cramer rule - for V_1 & V_2.

→ Excercise.

File No.

* Mesh analysis for AC circuit (example)



For loop 1, using KVL-

$$20\angle 0^\circ - I_1 Z_1 - (I_1 - I_2)\cdot Z_2 = 0$$

$$20\angle 0^\circ - I_1 (j12) - (I_1 - I_2)\cdot 15 = 0$$

$$j12\, I_1 + 15\, I_1 - 15\, I_2 = 20\angle 0^\circ$$

$$I_1 (15 + j12) - 15\, I_2 = 20\angle 0^\circ$$

$$19.209\angle 38.659^\circ\, I_1 - 15\, I_2 = 20\angle 0^\circ \quad -(1)$$

for loop 2, using KVL-

$$15(I_1 - I_2) - Z_3 I_2 - 40\angle -90^\circ = 0$$

$$15 I_1 - 15 I_2 - (-j8)\, I_2 = 40\angle -90^\circ$$

$$15 I_1 - \cancel{15}(15 - j8)\, I_2 = 40\angle -90^\circ$$

$$15\, I_1 - 17\angle -28.07^\circ\, I_2 = 40\angle -90^\circ \quad -(2)$$

$$\begin{bmatrix} 15+j12 & -15 \\ 15 & -15+j8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 40\angle -90^\circ \end{bmatrix}$$

$$\begin{bmatrix} 15+j12 & -15 \\ 15 & -15+j8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 20 \\ -40j \end{bmatrix}$$

using crammers rule -

$$\Delta = -96-60i = 113.207\angle -147.99$$

$$\Delta_1 = \begin{bmatrix} 20 & -15 \\ -40j & -15+j8 \end{bmatrix} = -300-440j = 532.54\angle -124.3$$

$$I_1 = \frac{\Delta_1}{\Delta} = 4.307 + 1.891j = 4.704\angle 23.703 \text{ A}$$

$\rightarrow 180 - 600j$

similarly, $\Delta_2 = \begin{bmatrix} 15+j12 & 20 \\ 15 & -40j \end{bmatrix} = \underline{180-60j} \times$

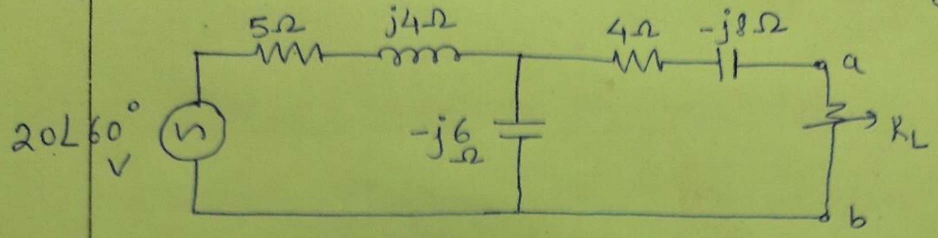
$= \boxed{189.74\angle -18.43^\circ \text{ A}} \times$

$626.41\angle -73.3^\circ \text{ A}$

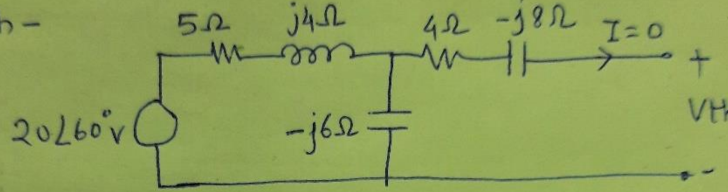
$$I_2 = \frac{\Delta_2}{\Delta} = \boxed{\frac{189.74\angle -18.43}{113.207\angle -147.99} = 1.676\angle 129.55^\circ \text{ A}} \times$$

$$I_2 = \Delta_2/\Delta = \frac{626.41\angle -73.3}{113.207\angle -147.99} = 5.53\angle 74.68^\circ \text{ A}$$

Find the Thevenin's equivalent for the following ckt



for Vth -



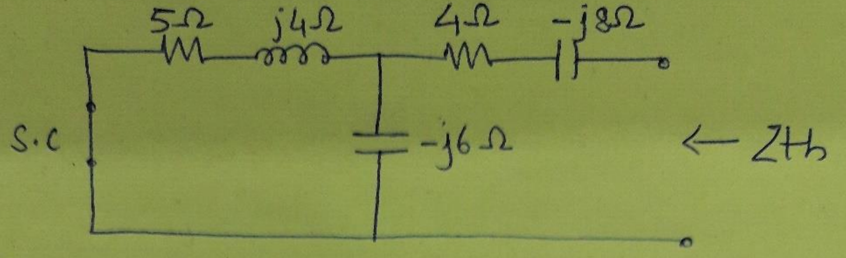
Vth = Voltage across $\underline{-j6\Omega}$.

using voltage division principle

$$V_{j6\Omega} = V_{th} = \frac{20\angle 60^\circ \times (-j6\Omega)}{5\Omega + j4\Omega - 6j\Omega}$$

$$V_{th} = 22.055 + (-3.177j) = \underline{\underline{22.283\angle -8.198^\circ\,V}}$$

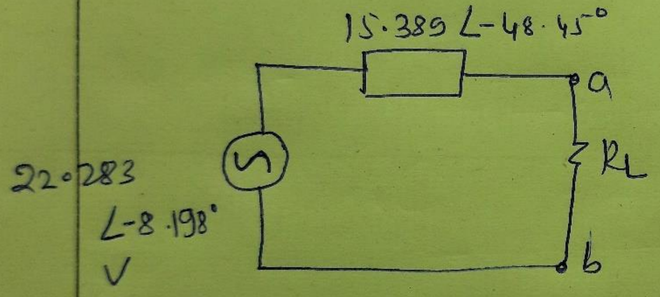
for Z_{th} -



$$Z_{th} = (5+j4)\Omega \parallel (-j6\Omega) + (4\Omega - j8\Omega)$$

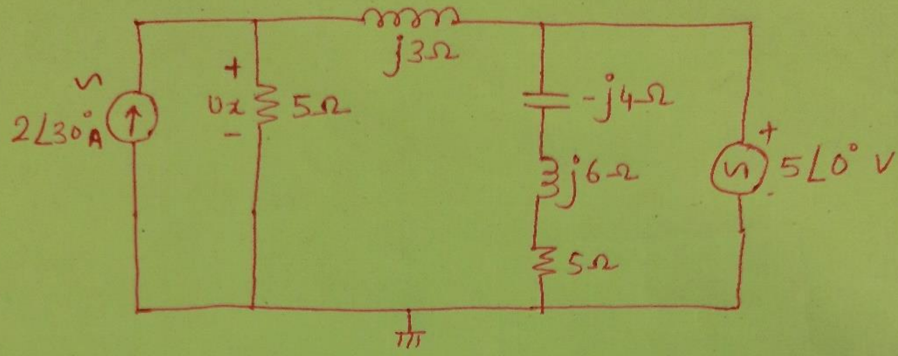
$$= 6.2068 - 3.517j + 4\Omega - j8\Omega$$

$$= 10.2068 - 11.517j = 15.389 \angle -48.45$$

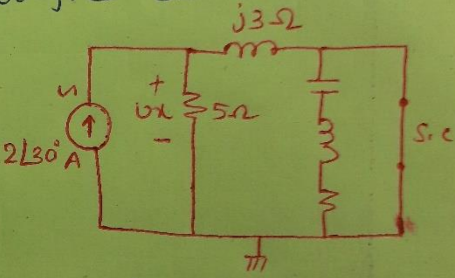


Superposition Theorem for AC Network:-

For the following network find voltage 'v_x' using superposition theorem.



Using current source alone. Modified ckt will be -



→ Using current division principle -

$$I'_{5\Omega} = \frac{2\angle 30° \times j3\Omega}{5+j3\Omega}$$

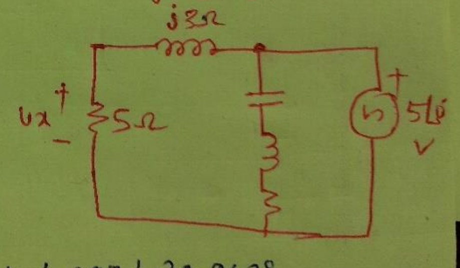
$= 0.0173 + 1.0288i$

$= 1.0289 \angle 89.03°\ A.$

$\therefore v'_x = 5 \times I_{5\Omega}$

$= 5.1445 \angle 89.03°\ V$

Considering voltage source alone. The modified ckt will be



using voltage division principle.

$$V''_x = \frac{5 \times 5\angle 0°}{5+j3} = 3.676 - 2.205i$$

$V''_x = 4.287 \angle -30.963°\ V$

$V_x = V'_x + V''_x = 5.1445\angle 89.03° + 4.287\angle -30.963°$

$= 3.763 + 2.938i = 4.774 \angle 37.98°\ V$